

Madi (Madeline) Abio

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EDUCATION

Griffith University

Bachelor's of Computer Science / Bachelor's of Electrical Engineering (Honours)

- Cumulative GPA: 6.32/7

Gold Coast, Queensland

Mar 2022 — Nov 2026

University of Utah

International Trimester Exchange

Salt Lake City, Utah, USA

2024

AWARDS & SCHOLARSHIPS

- Academic Excellence** | Griffith University, 2023-2025
- Chancellor's Scholarship** | Griffith University, 2025
- Amplify Connector** | Energy Queensland, 2025 – Present
- Future of Energy Scholarship** | Energy Queensland, 2023 – 2025
- Students in Power Bursary** | Australian Institute of Power, 2022 – 2023
- Brighter Futures Scholarship** | The Abedian Foundation, 2022

WORK EXPERIENCE

Founding Engineer

VPTech

Aug 2025 — Present

Gold Coast, QLD

- Own end-to-end feature delivery across web applications, backend services, AI agents and cloud infrastructure for an AI-native clinical platform.
- Built the company's first AI agent, a proof-of-concept that validated the product's flagship feature and set the architectural direction for the agent platform.
- Own code quality and maintainability, building and maintaining automated CI checks covering linting, security scanning, architectural boundaries and design-system consistency.
- Mentor engineers in code cleanliness and engineering best practices through code review and team-wide standards.
- Drive technology discovery by scouting and evaluating emerging tools for adoption.
- Lead sprint planning.
- Improved developer experience by introducing stacked PRs, on-demand preview environments and observability tooling for debugging agent behaviour.

Grid Technology Undergraduate Engineer

Energy Queensland

Jan 2025 — Sept 2025

Brisbane, QLD

- Developed an anomaly detection machine learning model to detect neutral faults targeting deployment across 10,000 smart meters (see: Neutral Fault Detection project).
- Streamlined daily operational data analysis by building a solar panel data dashboard in Python/Power BI, reducing manual processing time from 15 minutes to 15 seconds.
- Managed project delivery using Agile principles, including facilitating weekly sprints.
- Automated various non-technical workflows.

Undergraduate Research Assistant

Griffith University

Mar 2024 — Jun 2024

Gold Coast, QLD

- Performed data pre-processing for a machine learning research project by cleaning and structuring large datasets using Pandas (Python), Bash and the HDF5 format.

Cyber Security Platforms Intern

Energy Queensland

Nov 2023 — Feb 2024

Brisbane, QLD

- Developed a strong foundation in computer networking through self-directed training and mentorship, covering the OSI model, network protocols, and enterprise network architecture.
- Resolved ServiceNow tickets by applying network and security protocol knowledge.
- Gained exposure to enterprise cybersecurity practices, including cloud security, firewall management (Panorama), and Netskope.

PROJECTS

Automated First-Order Logic Theorem Prover

Rust, Apr 2026 – May 2026

- Implemented a backward sequent-calculus (LK') proof search engine for first-order logic in Rust.
- Extended it with a 6-class priority schedule ordering rule applications by productivity, and an iterative-deepening envelope around the depth-first kernel.
- Built a parallelized engine sweep and benchmark generator in Python.

Full-Stack LeetCode & Codeforces Web App (madiab.io)

TypeScript, March 2026

- Designed a full-stack web app hosting my LeetCode solve times and portfolio.
- NestJS backend, Next.js frontend, PostgreSQL database, Prisma ORM and Better Auth for authentication.
- Linked the projects section to my GitHub repositories via the public API.
- Hosted on Railway.

Digital Bass Synthesizer on Microcontroller

Embedded C, Sept 2025 – Oct 2025

- Developed bare-metal C firmware to generate waveforms, control ADSR envelopes and handle input from potentiometers (via ADC) and a matrix keypad.
- Engineered a custom external clock circuit to bridge the I²C peripheral for compatibility with the I²S DAC.
- Drove the 240x320 LCD over SPI for waveform and real-time ADSR visualization.
- Implemented DMA ping-pong buffers for continuous audio streaming and optimized ISR timing for stable waveform output.

Neutral Fault Detection (Anomaly Detection)

Python, Feb 2025 – Sept 2025

- Developed an in-house neutral fault detection algorithm to investigate excessive false positives in a third-party solution.
- Identified systematic false positives on solar PV sites, caused by violated voltage-current modelling assumptions making normal solar behaviour resemble fault signatures.
- Proposed a revised approach that explicitly accounted for solar PV electrical behaviour.
- Designed and prototyped a scalable architecture targeting deployment across 10,000+ smart meters.
- Built a parallelized feature extraction and evaluation pipeline with time-ordered, meter-isolated validation and failure-mode analysis.

Encryption Client

C++, May 2025

- Developed a terminal-based multi-round encryption/decryption client, applying OOP principles (abstract base classes, inheritance, polymorphism) for modular, reusable design.
- Implemented a finite-state-machine menu with input validation and exception handling, using the STL and lambdas for efficient message processing.

CLUBS & SOCIETIES

- **Ravi's Study Program** | *Mentee, 2026*
- **Griffith University Coding Club (GCC)** | *ICPC Div B Competitor & Member, 2025 – Present*
- **UQ Computing Society** | *Lead Guitar Player (UQCS Band), Competitive Programmer, 2025 – Present*
- **UQ Fintech Society** | *Member, 2025 – Present*
- **Griffith University Advanced Robotics Development (GUARD)** | *Member, 2023 – Present*
- **Griffith University Gold Coast IEEE Student Branch** | *Administrator, 2023*

SKILLS

- **Programming Languages:** Bash, C, C++, C#, Python, Rust, TypeScript
- **Software Development & Tools:** Agentic Software Development (with Claude Code), AWS, CI/CD, Drizzle, Fastify, Git, Linux, NestJS, Next.js, Object-Oriented Programming, oRPC, PostgreSQL, Prisma, React, Test-Driven Development
- **Machine Learning & Data Science:** Applied Machine Learning, Data Science, h5py, NumPy, Pandas, scikit-learn
- **LLMOps:** AgentCore, LangChain, LangGraph, LLM orchestration
- **Embedded & FPGA:** Embedded/Bare Metal C, PCB Design, SystemVerilog
- **Professional Skills:** Communication, Problem Solving, Project Management, Teamwork, Technical Writing